

Unconditional Market-Realism Evaluation

Coinbase BTC / ETH / SOL — free-running simulation vs. real order flow, 2026-07-15

1. Protocol

Ten metric families measure how closely *free-running* simulated order flow matches real market data, without conditioning on trading actions. Every model is evaluated with identical code on its **existing calibrated rollouts** ($32 \times 600\text{s} \times 3$ rollout seeds per checkpoint, reproduced exactly from banked calibration constants); real data is scored on **equal-duration bootstrap segments** of matching count. Statistics: rollout seeds averaged within checkpoint, 95% *t*-CIs across checkpoints; **bold** = best per row (SAHP excluded from bolding: uncalibrated clock, †). Checkpoints that failed calibration verification are excluded (s2P2: 2/3 on BTC and ETH, 3/3 on SOL — its cells are $n=1$ or absent).

Book-state metrics (spread, imbalance) replay both streams through the same deterministic book engine with a shared depth profile, so book assumptions cancel in the comparison. Mid-price returns and price-change times use the signed-order-flow proxy price (the paper’s return construction): type-only, hence symmetric across real and simulated streams and responsive at tick-constrained books where a volume-free replayed mid barely moves — for the same reason, the *spread* rows carry little discriminative power on these assets (near-constant one-tick spreads for real and simulated flow alike) and are reported for completeness only.

2. Cross-asset summary (mean rank over the 14 distance rows)

model	BTC	ETH	SOL
NHP	2.93	2.00	2.43
LSTM	3.93	4.00	3.57
SAHP †	3.50	4.07	3.71
PCT-LSTM	3.07	3.14	2.14
s2P2	4.29	4.93	—
ss2P2	3.29	2.86	3.14

No model sweeps the battery. NHP leads the mean rank on BTC/ETH via the sequential-composition families (transitions, marginals); ss2P2 leads the temporal and price-path families (inter-event W_1 , return W_1 , price-change times, Fano) — the quantities a closed-loop simulator exists to reproduce; PCT-LSTM leads SOL.

3. Per-asset distance tables

BTC (38 ev/s)

distance to real	NHP	LSTM	SAHP †	PCT-LSTM	s2P2	ss2P2
event types (JS)	0.062±0.057	0.110±0.109	0.071±0.150	0.056±0.060	0.319	0.193±1.033
event types (TV)	0.289±0.124	0.377±0.233	0.301±0.375	0.255±0.089	0.695	0.488±1.671
inter-event Δt (KS)	0.298±0.027	0.418±0.014	0.528±0.195	0.350±0.028	0.109	0.118±0.071
inter-event Δt (W_1 , s)	0.018±0.004	0.028±0.001	0.092±0.176	0.020±0.002	0.031	0.011±0.028
transitions (Frobenius)	1.735±0.078	2.057±0.521	1.723±0.624	1.842±0.538	1.995	2.407±6.708
transitions (row KL)	0.297±0.059	0.511±0.263	0.363±0.256	0.310±0.175	0.311	0.471±2.006
level marginal (JS)	0.023±0.059	0.024±0.040	0.022±0.054	0.026±0.044	0.303	0.161±1.007
spread (KS)	0.001±0.000	0.001±0.000	0.026±0.104	0.016±0.051	0.001	0.001±0.000
spread (W_1 , ticks)	0.00±0.00	0.00±0.00	0.03±0.14	0.02±0.05	0.00	0.00±0.00
imbalance (KS)	0.614±0.183	0.533±0.171	0.328±0.193	0.350±0.374	0.589	0.580±3.228
1 s mid returns (KS)	0.263±0.368	0.517±0.466	0.136±0.087	0.239±0.081	0.170	0.188±0.632
1 s mid returns (W_1 , ticks)	5.83±8.72	5.71±7.08	2.99±2.74	2.37±1.73	2.61	1.83±1.30
price-change Δt (KS)	0.230±0.022	0.300±0.344	0.410±0.592	0.302±0.235	0.304	0.224±0.335
Fano 1–100 s (rel-err)	0.867±0.067	0.993±0.000	0.726±0.225	0.776±0.154	1.710	0.757±1.407

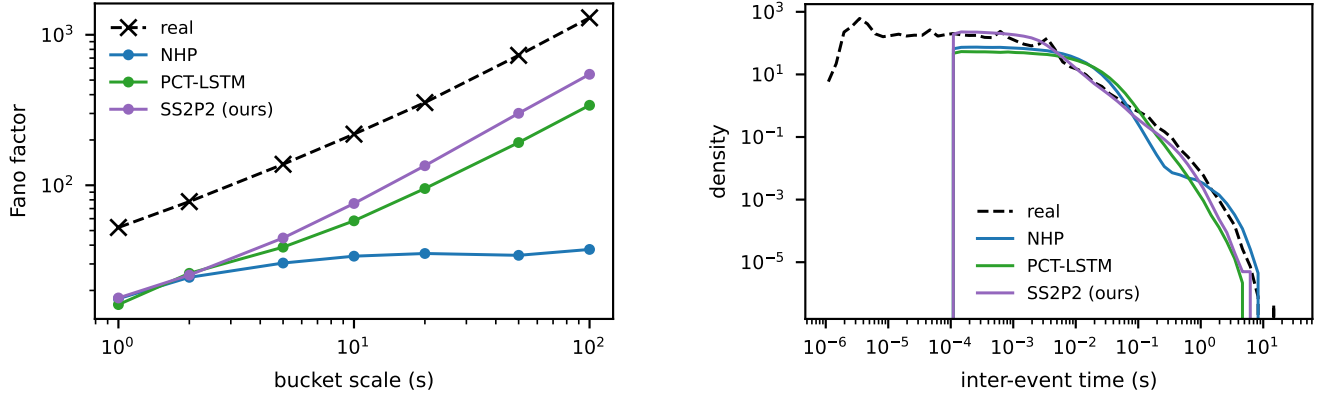
ETH (48 ev/s)

distance to real	NHP	LSTM	SAHP†	PCT-LSTM	s2p2	ss2p2
event types (JS)	0.033±0.036	0.059±0.040	0.115±0.075	0.048±0.077	0.176	0.088±0.024
event types (TV)	0.209±0.122	0.280±0.138	0.402±0.079	0.253±0.257	0.506	0.341±0.111
inter-event Δt (KS)	0.318±0.009	0.427±0.004	0.573±0.058	0.339±0.031	0.341	0.097±0.099
inter-event Δt (W_1 , s)	0.023±0.000	0.031±0.000	0.109±0.078	0.025±0.005	0.052	0.015±0.003
transitions (Frobenius)	1.286±0.376	1.549±0.253	1.571±0.408	1.517±1.069	2.230	1.322±0.190
transitions (row KL)	0.168±0.089	0.365±0.087	0.311±0.025	0.245±0.234	0.481	0.315±0.133
level marginal (JS)	0.005±0.004	0.013±0.027	0.029±0.003	0.007±0.017	0.103	0.041±0.044
spread (KS)	0.000±0.000	0.000±0.000	0.019±0.057	0.000±0.001	0.048	0.000±0.000
spread (W_1 , ticks)	0.00±0.00	0.00±0.00	0.02±0.06	0.00±0.00	89.19	0.00±0.00
imbalance (KS)	0.423±0.567	0.498±0.467	0.343±0.343	0.534±0.642	0.663	0.689±0.277
1 s mid returns (KS)	0.369±0.261	0.426±0.419	0.286±0.109	0.423±0.701	0.297	0.358±0.278
1 s mid returns (W_1 , ticks)	9.97±5.84	5.58±4.92	3.82±0.72	8.40±12.98	4.10	4.54±3.90
price-change Δt (KS)	0.178±0.009	0.392±0.094	0.477±0.193	0.223±0.015	0.218	0.175±0.059
Fano 1–100 s (rel-err)	0.864±0.113	0.995±0.000	0.893±0.159	0.725±0.227	7.881	0.781±0.035

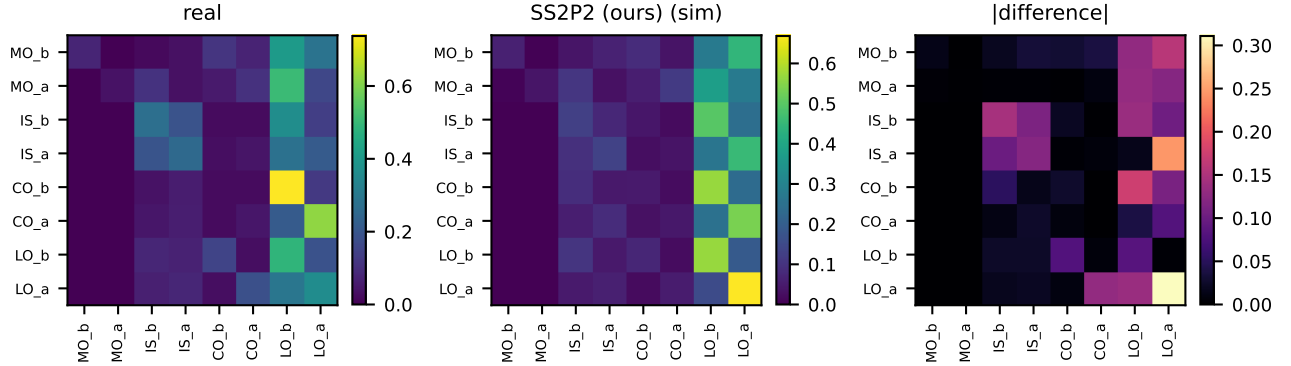
SOL (24 ev/s)

distance to real	NHP	LSTM	SAHP†	PCT-LSTM	s2p2	ss2p2
event types (JS)	0.021±0.025	0.034±0.056	0.030±0.052	0.026±0.012	–	0.088±0.213
event types (TV)	0.164±0.131	0.192±0.091	0.188±0.193	0.182±0.045	–	0.315±0.405
inter-event Δt (KS)	0.379±0.018	0.497±0.003	0.580±0.041	0.435±0.014	–	0.168±0.043
inter-event Δt (W_1 , s)	0.032±0.010	0.044±0.000	0.108±0.062	0.031±0.006	–	0.018±0.005
transitions (Frobenius)	0.805±0.094	0.919±0.163	0.886±0.387	1.046±0.037	–	1.322±1.532
transitions (row KL)	0.113±0.047	0.205±0.187	0.150±0.185	0.191±0.065	–	0.223±0.340
level marginal (JS)	0.014±0.026	0.007±0.004	0.021±0.051	0.013±0.007	–	0.058±0.161
spread (KS)	0.018±0.057	0.008±0.004	0.032±0.059	0.007±0.004	–	0.008±0.003
spread (W_1 , ticks)	0.02±0.06	0.01±0.00	0.04±0.08	0.01±0.00	–	0.01±0.00
imbalance (KS)	0.542±0.547	0.608±0.335	0.323±0.306	0.397±0.229	–	0.448±0.612
1 s mid returns (KS)	0.054±0.046	0.273±0.021	0.113±0.005	0.112±0.210	–	0.051±0.038
1 s mid returns (W_1 , ticks)	0.23±0.10	0.69±0.48	0.38±0.05	0.32±0.61	–	0.21±0.14
price-change Δt (KS)	0.329±0.065	0.318±0.140	0.526±0.168	0.278±0.032	–	0.335±0.203
Fano 1–100 s (rel-err)	0.749±0.172	0.988±0.000	0.917±0.091	0.448±0.057	–	0.630±0.391

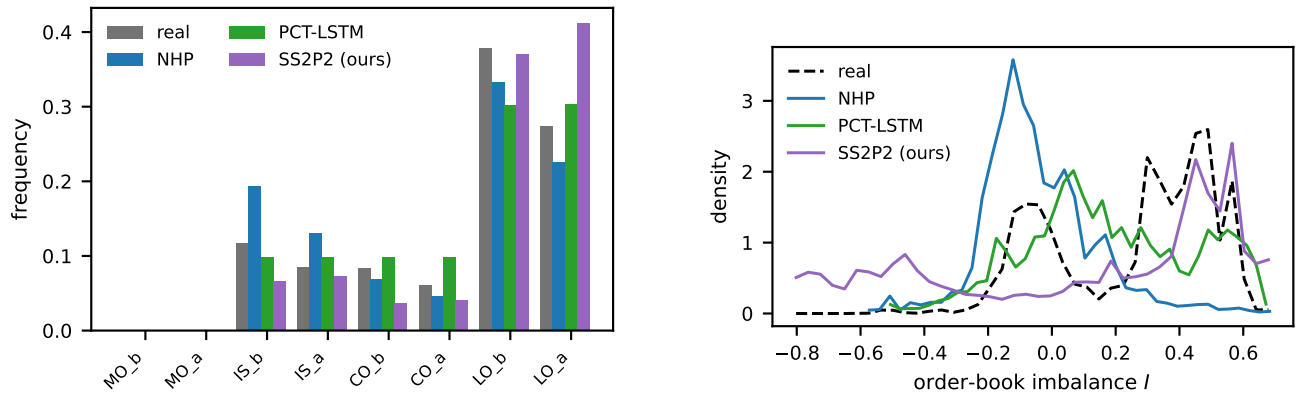
4. Figures (BTC; representative rollout for density/QQ panels)



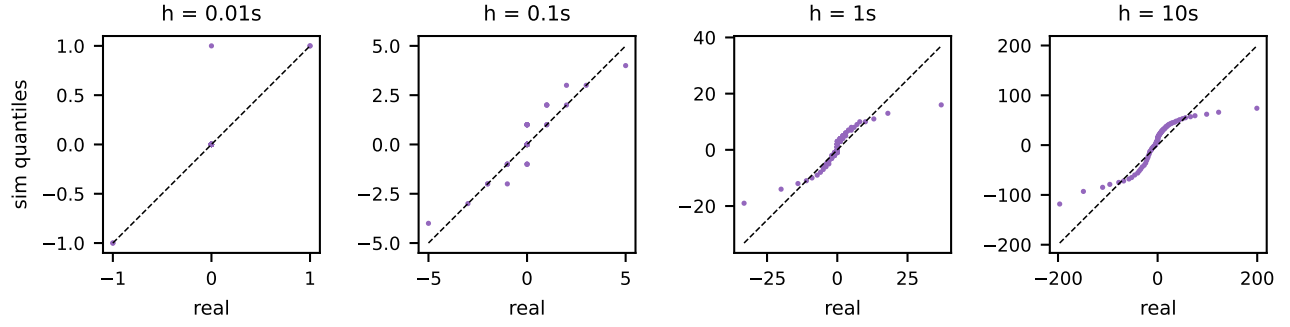
Left: Fano factor of event counts, 1–100 s. Real order flow (dashed) is super-Poissonian with Fano rising across two decades of scale; NHP’s clustering *saturates* beyond 5 s while SS2P2 tracks the rise — long-scale burst structure only carried-state simulation reproduces. **Right:** inter-event-time densities (log-log); the state-space family matches the heavy right tail.



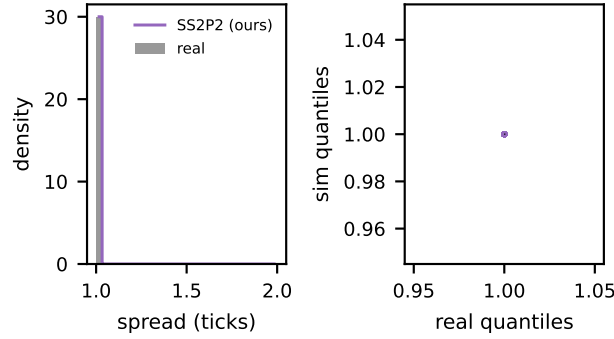
First-order type-transition matrices (coarse class $\times$ side groups): real, ss2p2 simulated, and absolute difference.



Left: coarse event-group frequencies. **Right:** order-book imbalance densities from the shared-book replay.



Proxy mid-return QQ plots at 10 ms–10 s horizons (ss2p2 vs. real).



Spread distribution and QQ (reported for completeness; see the protocol caveat).

5. Reading

ss2p2 wins where a simulator earns its keep. On BTC it takes inter-event W_1 (0.011s vs. NHP’s 0.018), 1s-return W_1 (1.83 ticks vs. 5.83 — $3\times$ closer), price-change-time KS, and the extended Fano error; the Fano figure shows why: its simulated burst hierarchy keeps compounding to 100s where NHP flattens. **nhp wins next-event composition** (transition Frobenius/KL, level marginals), and PCT-LSTM the event-type marginals — consistent with the prediction tables. **s2p2 is barely measurable:** most of its cells rest on a single surviving checkpoint per asset (none on SOL), and where present it trails on marginal and Fano families — the realism battery inherits the calibratability findings. All numbers regenerate with three commands (`paper/scripts/README.md`); streams are banked, so metric iterations are CPU-only.